

# Aliah University

Department of Electrical Engineering, B-Tech

End-Semester Examination, Spring-2022

Subject: Basic Electrical Engineering [EENUGES01]

Full Marks: 80

Year: I

Time: 3 Hours

Semester: II

Answer any five questions

5x16=80 Marks

1. a) Define i) charge ii) electric current iii) power iv) network 4 Marks
- b) State the steps to solve an electric circuit problem by the Thevenin's Theorem. 3 Marks
- c) For the circuit shown in Fig. 1 find  $V_{CE}$  and  $V_{AG}$  4 Marks

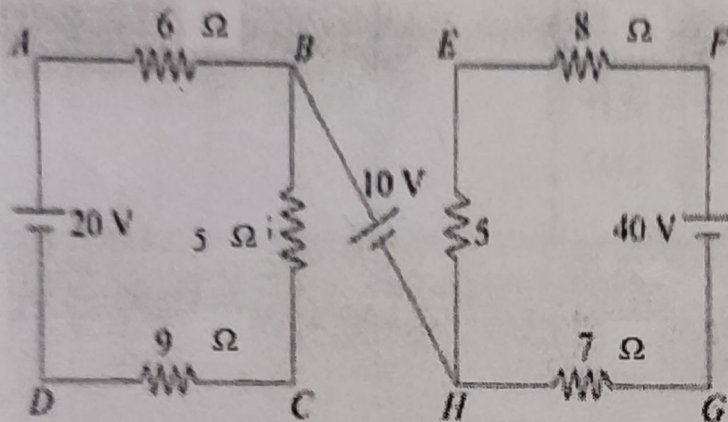


Fig. 1

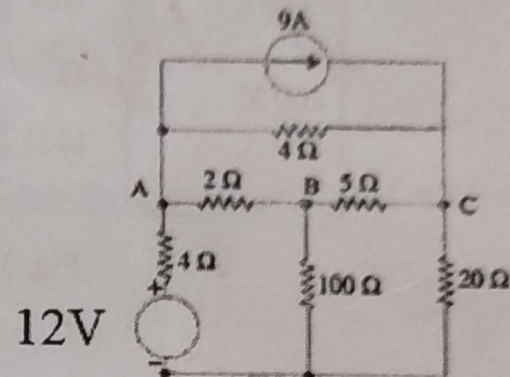


Fig. 2

- d) Use nodal analysis to determine the voltage across 5 ohm resistance and the current in the 12V source. (Refer. to Fig. 2) 5 Marks
2. a) What is an ideal transformer? Derive the condition of maximum efficiency for a single phase transformer using common notations. 5 Marks
- b) In a 50 kVA 2000/200V, 50 Hz single phase transformer, the iron and copper losses are found to be 800 W and 400 W at the full load and unity p.f, respectively. Calculate the efficiency of the transformer with same power factor at i) full load and ii) one third of load. 6 Marks
- c) Draw the phasor diagrams for different type of loads in an ideal transformer. 5 Marks
3. a) Define phasor and phase angle. Explain different phasor for a single phase circuit with relevant diagrams. 6 Marks
- b) An alternating voltage is given by  $V=230\sin 314t$ . Calculate i) frequency, ii) maximum value, iii) average value, iv) RMS value. 4 Marks
- b)** A series RLC circuit has  $25\Omega$  resistance,  $0.2\text{ mH}$  inductance and  $66.3\mu\text{F}$  capacitance and is connected to a 240V, 60Hz supply. Find (i) the current, (ii) the power factor and (iii) the voltage drop across each element of the circuit. 6 Marks
4. a) Draw the connection diagrams of D.C Generator for different type of excitations. 5 Marks
- b) A DC long shunt compound generator delivers a load current of 10A at 420V. The resistance of the armature, series field and shunt field are  $0.03\Omega$ ,  $0.015\Omega$  and  $150\Omega$ , respectively. Calculate the emf induced in the armature. Allow a brush voltage drop of 1V/brush. 6 Marks
- c) Sketch the O.C.C of D.C shunt generator. 2 Marks
- d) What is the significance of back EMF in a DC motor? 3 Marks
5. a) Derive the condition for maximum power transfer from a source to resistive load. 5 Marks
- b) What is DC motor? Derive the expression of torque for a DC motor. 6 Marks
- c) Write a short note on series resonance. 5 Marks



6. a) State Norton's theorem and find current using Norton's theorem through a load of 8 ohm in the circuit shown in Fig. 3 2x8=16 Marks
- b) In the network shown in Fig. 4, find the value of R such that maximum possible power will be transferred to  $R_L$ . Find also the value of the maximum power and the power supplied by source under this conditions.

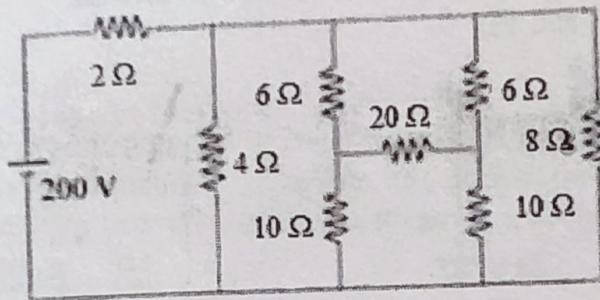


Fig. 3

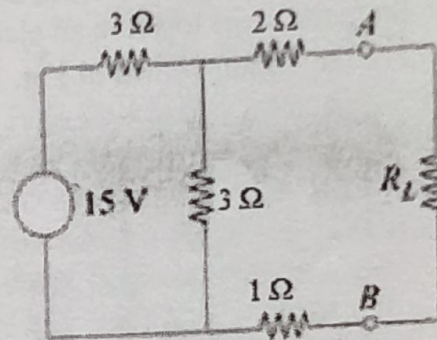


Fig. 4